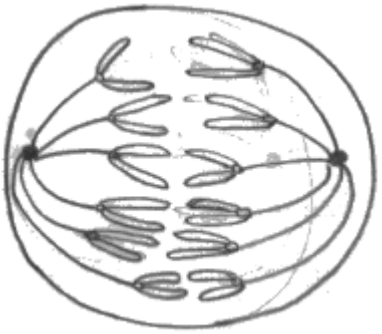


| Question Number | Answer                                                                                                                                                                                                        | Additional guidance                                                                                            | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------|
| <b>4(a)(i)</b>  | <p>An answer that makes reference to the following points:</p> <ul style="list-style-type: none"> <li>centrioles and spindle fibres shown (1)</li> <li>6 chromosomes being pulled to each pole (1)</li> </ul> | <p>Example of diagram:</p>  | <b>(2)</b> |

| Question Number | Answer                                                                                                                                                                                          | Additional guidance                                                                                                                           | Mark       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>4(a)(ii)</b> | <p>A calculation in which:</p> <ul style="list-style-type: none"> <li>calculation of number of degrees per minute (1)</li> <li>calculation of number of degrees (including unit) (1)</li> </ul> | <p><u>Example of calculation:</u></p> $270 \div (9 \times 60) = 0.5^\circ$ $0.5 \times 20 = 10^\circ$ <p>Correct answer scores full marks</p> | <b>(2)</b> |

| Question Number | Answer                                                                                                  | Additional guidance                                                                 | Mark       |
|-----------------|---------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------|
| <b>4(b)(i)</b>  | <ul style="list-style-type: none"> <li>location of gene(s) on a(n Indian muntjac) chromosome</li> </ul> | ACCEPT {nucleotide base pairs / allele} for gene<br>ACCEPT chromatid for chromosome | <b>(1)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                                                                         | Additional guidance                                                                                                                                                          | Mark       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>4(b)(ii)</b> | An explanation that makes reference to the following points: <ul style="list-style-type: none"> <li>(egg cells are) haploid (1)</li> <li>(egg cell chromosomes) have {an altered base sequence / different alleles} (than body cell chromosomes) (1)</li> <li>due to {(random) mutations (during DNA replication) / crossing over / random assortment / independent assortment} (1)</li> </ul> | ACCEPT contain {3 / half the number of} chromosomes<br><br>do not accept genes<br><br>ACCEPT chromosomes mutations due to errors in separation of {chromatids / chromosomes} | <b>(3)</b> |

| Question Number | Answer                                                                                                                                                                                                                                                                                                                                   | Additional guidance                                                                                                                                                            | Mark       |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <b>4(c)</b>     | <p>An answer that makes reference to two of the following points:</p> <ul style="list-style-type: none"> <li>• (because the parents are) different species (1)</li> <li>• (because {maternal and paternal / 3 and 23}) chromosomes would not pair up (1)</li> <li>• (therefore) cannot make {haploid / sperm / egg} cells (1)</li> </ul> | <p>ACCEPT because diploid number (of either species) could not be restored<br/>ignore parents have different number of chromosome pairs</p> <p>ACCEPT meiosis cannot occur</p> | <b>(2)</b> |